

Hotel Booking and Management System

A Database Management System

By Group (5)

Norah Eyad Alhindi (SID: 4450271)

Shahad Alhomaidi Alrashedi (SID: 4453301)

Lama Fayez Albeladi (SID: 4453423)

Zainab Hussain Alnakhli (SID: 4453753)

A Project Report Submitted to Fulfil the Requirements of CS372
Supervised by: Dr. Lyakothonisa

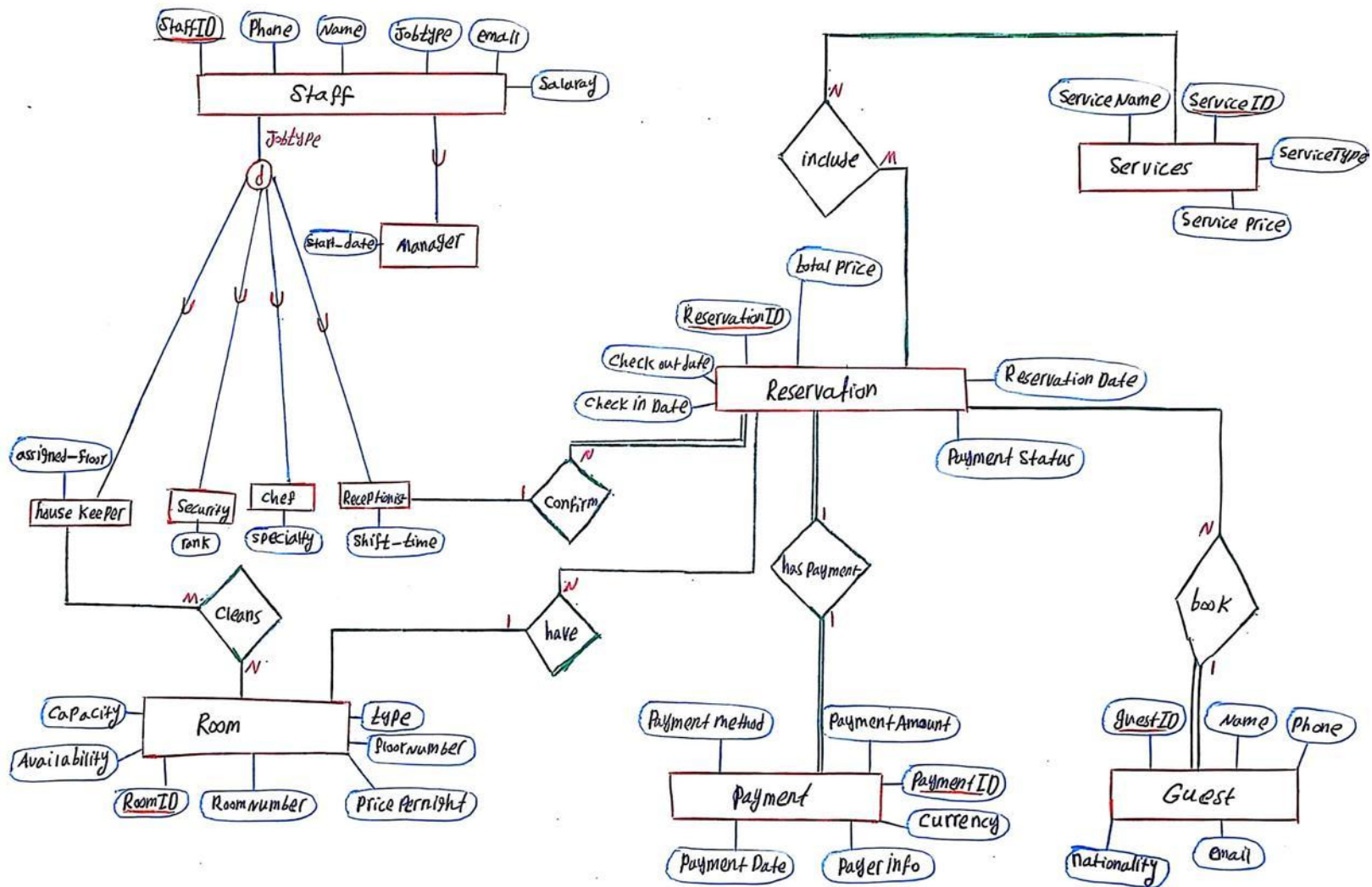
1. Introduction

Many organizations rely on manually recorded data or scattered digital files, which leads to inconsistent information, duplication, and difficulty tracking updates. We introduced The Hotel Booking and Management System, one that addressed many factors that needed to be studied in order to measure the company's success. This domain depends heavily on structured data, for example the number of reservations in a year can help determine if it's on the right track or not, having a structured dataset means that it could help you identify what attributes are contributing to the success of your business and what are affecting it negatively. Hotels that don't pay attention to details like this and work only on poor data handling could drive their business's downhill into failure due to lack of structure.

We chose this problem domain because we believe if the right tools are studied better outcomes arise. The main goal of our solution is to introduce efficiency and accuracy helping hotel owners acknowledge their strengths and weaknesses by simply providing a reliable way to store and manage such data. Organizing scattered information into an arranged, centralized database and reducing human error by automating data storage and retrieval supports the long-term growth of the system.

Our Database Management System will consider the following entities: Staff, Services, Reservations, Guests, Rooms, and Payments with contributing attributes like salaries, prices, availability, dates, and many more factors. By using SQL, our system ensures consistent data storage, eliminates duplication, maintains relationships between entities, and provides fast access to information through optimized queries.

2. Conceptual Model Design (EER Diagram)



3. Mapping Conceptual Model Design to Relational Model Design

- 1) For each regular(strong) entity type E in the EER schema, we create a relation R that includes all the simple attributes of that entity and choose a key attribute for each relation.

Staff					
<u>StaffID</u>	PHONE	Name	JobType	email	Salary

Reservation					
<u>ReservationID</u>	Total price	Reservation Date	check out date	check in date	payment status

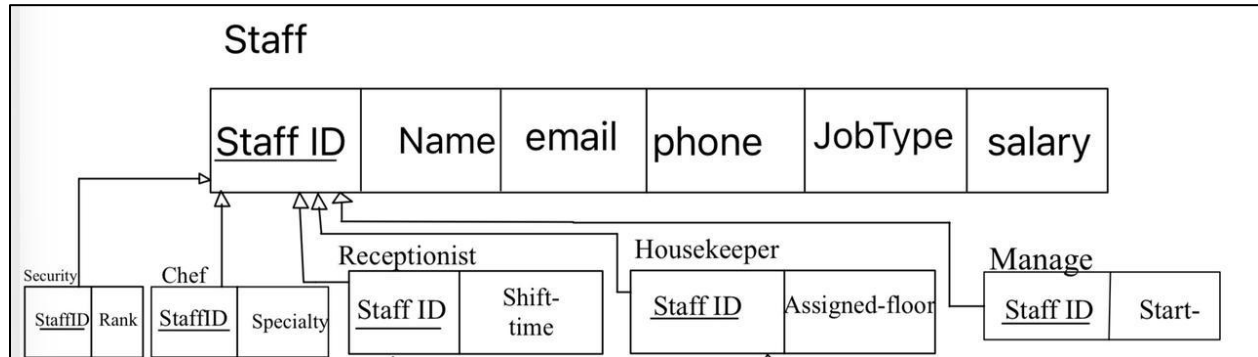
Payment					
<u>PaymentID</u>	Payment date	Payment method	Payer info	Payment amount	Currency

Services			
<u>ServiceID</u>	Service name	Service price	Service Type

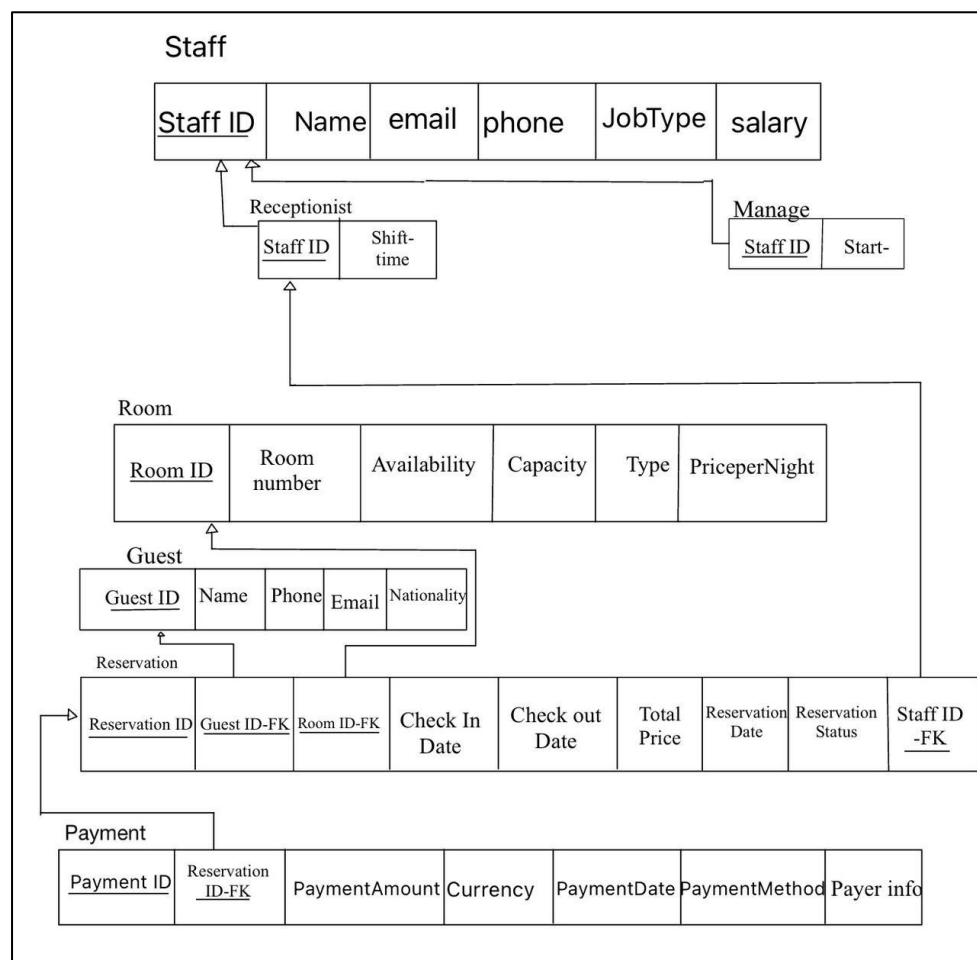
Guest				
<u>GuestID</u>	Name	Phone	email	nationality

Room						
<u>RoomID</u>	Type	Floor number	Price per night	Capacity	Availability	Room Number

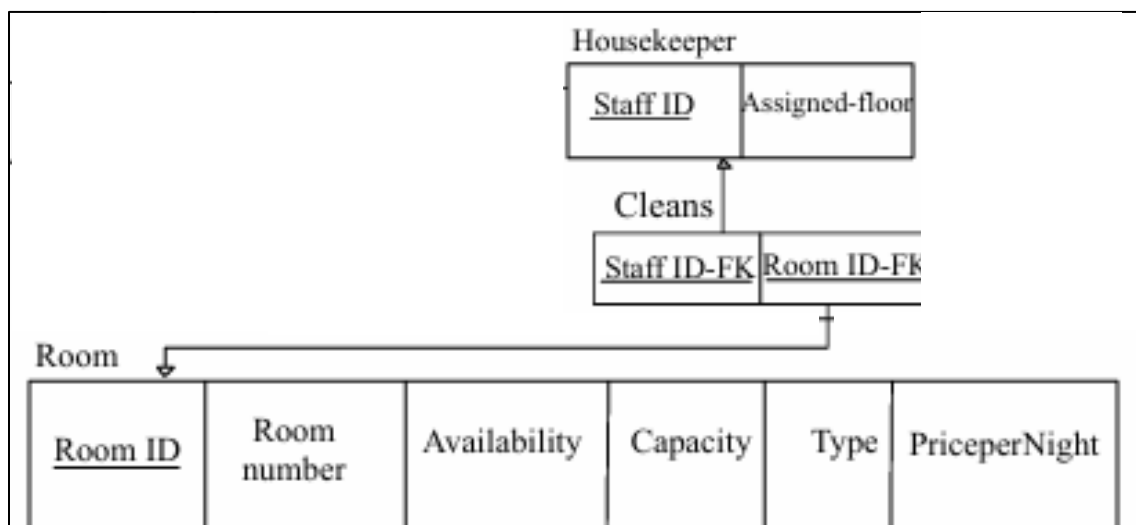
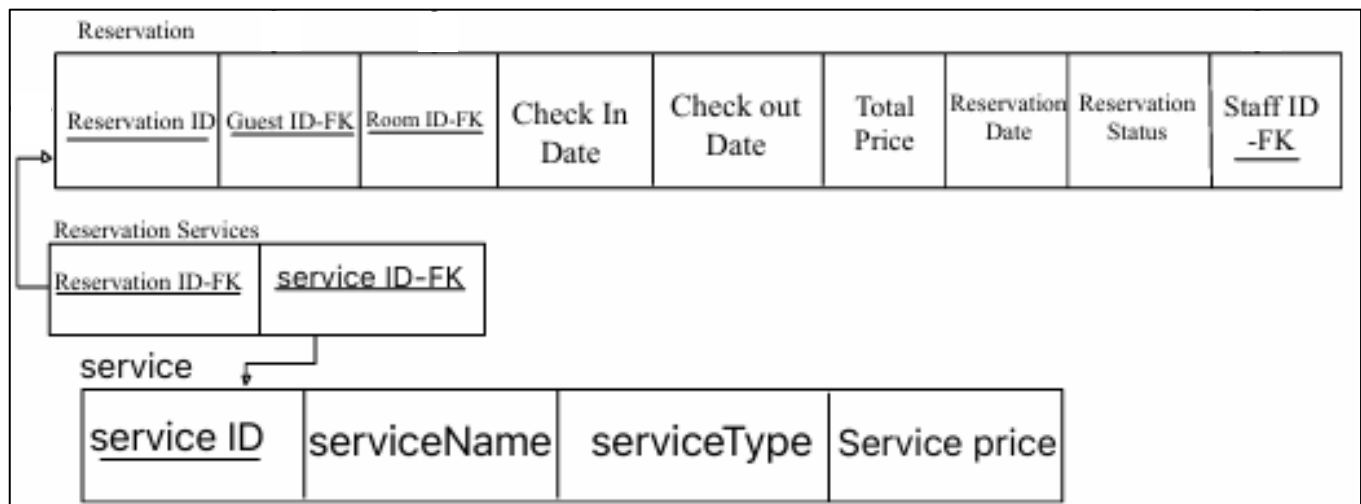
- 2) For each specialization, a relation is created for the superclass and a relation for each subclass. The primary key of the superclass is used as a foreign key in each subclass relation.



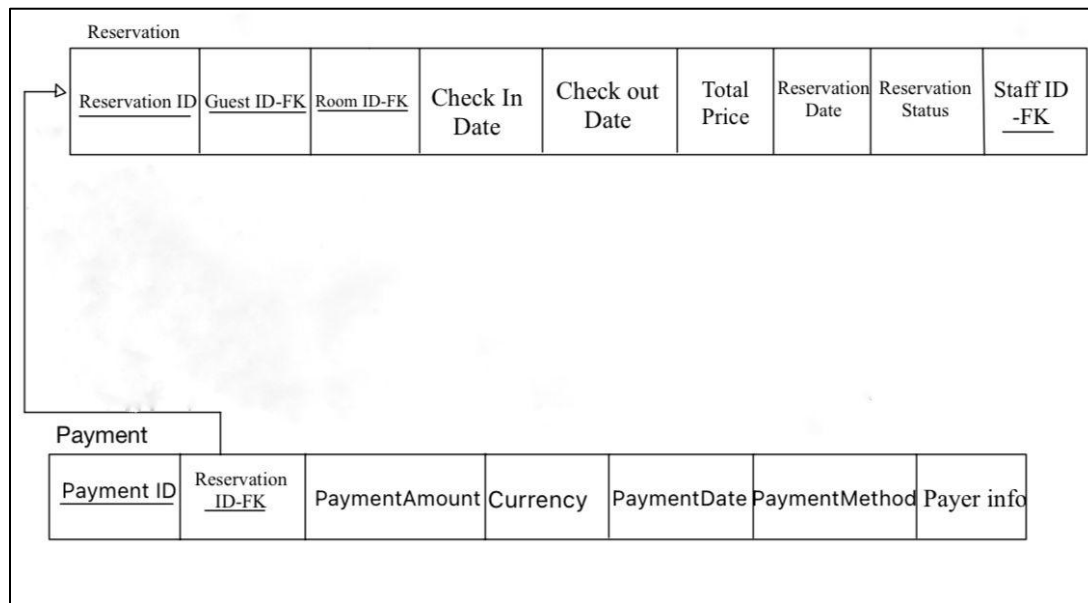
- 3) For each 1–N relation, a relation is created for the entity on the side representing the large number many. The primary key of the entity on the side representing one one is used as the foreign key in the relation for the large side, and any properties or attributes of the relation are included in that relation.



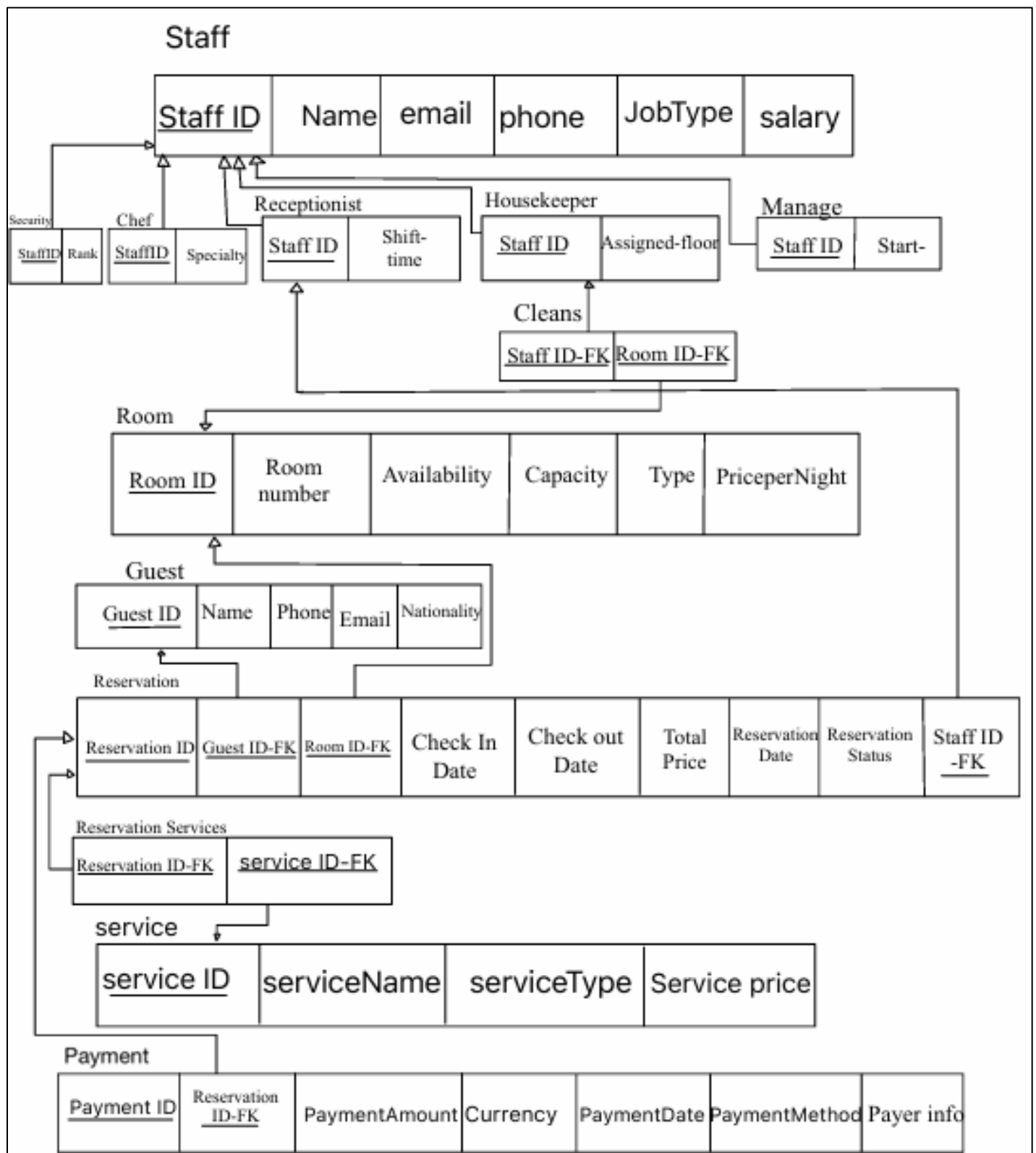
- 4) For each M-N relationship, we created a separate relation to represent the relationship itself. The primary keys of the participating entities were included in this relation as foreign keys, and they form the primary key of the new relation. And any attributes that belong to the relationship were also included.



- 5) For a 1–1 relationship, a foreign key is added to one of the participating entities. The primary key of the other entity is used as a foreign key in the entity with total participation.



3. Relational Model Design



4.

<pre>SELECT Receptionist.Shift_Time, Staff.Name, Staff.Phone, Staff.Email FROM Receptionist RIGHT OUTER JOIN Staff ON Receptionist.Staffid = Staff.StaffID ;</pre>			
Query Result x			
SQL All Rows Fetched: 15 in 0.018 seconds			
SHIFT_TIME	NAME	PHONE	EMAIL
1	1 Sara	551000001	sara@hotel.com
2	0 Omar	551000002	omar@hotel.com
3	1 Rana	551000003	rana@hotel.com
4	1 Noor	551000004	noor@hotel.com
5	0 Alya	551000005	alya@hotel.com
6	(null) Mona	551000007	mona@hotel.com
7	(null) Lyan	551000013	lyan@hotel.com
8	(null) Rita	551000008	rita@hotel.com
9	(null) Salem	551000009	salem@hotel.com
10	(null) Areej	551000010	areej@hotel.com
11	(null) Jalal	551000015	jalal@hotel.com
12	(null) Majed	551000012	majed@hotel.com
13	(null) Huda	551000011	huda@hotel.com
14	(null) Layla	551000014	layla@hotel.com
15	(null) Fahad	551000006	fahad@hotel.com

```
SELECT ServiceType, COUNT(ServiceName) AS TotalServices FROM Services GROUP BY ServiceType;
```

Query Result x

SQL | All Rows Fetched: 2 in 0.007 seconds

SERVICETYPE	TOTALSERVICES
1 Food and Beverage Services	2
2 Leisure and Wellness Services	3

```
SELECT * FROM Manager LEFT OUTER JOIN Staff
ON Staff.StaffID = Manager.StaffID
ORDER BY Start_date DESC;
```

Query Result x

SQL | All Rows Fetched: 5 in 0.016 seconds

	STAFFID	START_DATE	STAFFID_1	NAME	EMAIL	PHONE	JOBTYPE	SALARY
1	115	05-MAR-24	115	Jalal	jalal@hotel.com	551000015	Manager	7900
2	113	10-SEP-23	113	Lyan	lyan@hotel.com	551000013	Manager	8100
3	111	01-JAN-23	111	Huda	huda@hotel.com	551000011	Manager	8000
4	112	15-JUN-22	112	Majed	majed@hotel.com	551000012	Manager	8200
5	114	20-NOV-21	114	Layla	layla@hotel.com	551000014	Manager	8300

```
SELECT Staff.Name, Staff.StaffID, Staff.JobType FROM Staff INNER JOIN Housekeeper
ON Staff.StaffID = Housekeeper.StaffID ORDER BY Name ASC ;
```

Query Result x

SQL | All Rows Fetched: 5 in 0.043 seconds

	NAME	STAFFID	JOBTYPE
1	Areej	110	Housekeeper
2	Fahad	106	Housekeeper
3	Mona	107	Housekeeper
4	Rita	108	Housekeeper
5	Salem	109	Housekeeper

<pre> SELECT GuestID, COUNT(*) AS TotalReservations FROM Reservation GROUP BY GuestID ORDER BY TotalReservations DESC; </pre>		
Query Result x SQL All Rows Fetched: 5 in 0.038 seconds		
	GUESTID	TOTALRESERVATIONS
1	204	2
2	205	2
3	202	1
4	201	1
5	203	1

```
SELECT *  
FROM Room  
WHERE Availability = 'Available';
```

Query Result x

SQL | All Rows Fetched: 2 in 0.006 seconds

	ROOMID	ROOMNUMBER	AVAILABILITY	CAPACITY	TYPE	PRICEPERNIGHT
1	1	101	Available	2	Standard	250
2	3	201	Available	4	Suite	750

```
SELECT RoomNumber, Type, PriceperNight  
FROM Room  
WHERE PriceperNight < 300;
```

Query Result x

SQL

| All Rows Fetched: 2 in 0.006 seconds

ROOMNUMBER

TYPE

PRICEPERNIGHT

1

101 Standard

250

2

202 Standard

200

```

SELECT *
FROM Payment
WHERE PaymentMethod = 'Credit Card';

```

Query Result x

SQL | All Rows Fetched: 2 in 0.046 seconds

	PAYMENTID	RESERVATIONID_FK	PAYMENTAMOUNT	CURRENCY	PAYMENTDATE	PAYMENTMETHOD	PAYERINFO
1	501	1001	1250 SAR		10-JAN-24	Credit Card	Ahmad - Visa **** 1122
2	504	1006	1200 SAR		06-JUN-24	Credit Card	Ahmad - Mastercard **** 7741

```

SELECT GuestID,RoomID,ReservationDate
FROM Reservation
ORDER BY ReservationDate ASC;

```

Query Result x

SQL | All Rows Fetched: 7 in 0.045 seconds

	GUESTID	ROOMID	RESERVATIONDATE
1	202	2	01-JAN-24
2	201	1	02-JAN-24
3	203	3	05-MAR-24
4	204	4	05-APR-24
5	205	1	20-APR-24
6	205	1	20-AUG-24
7	204	2	15-SEP-24

```
SELECT *
FROM Payment;
```

Query Result x

SQL | All Rows Fetched: 4 in 0.002 seconds

	⚡ PAYMENTID	⚡ RESERVATIONID_FK	⚡ PAYMENTAMOUNT	⚡ CURRENCY	⚡ PAYMENTDATE	⚡ PAYMENTMETHOD	⚡ PAYERINFO
1	501	1001	1250 SAR		10-JAN-24	Credit Card	Ahmad - Visa **** 1122
2	502	1003	1500 SAR		09-MAR-24	Cash	Lama - Paid at Reception
3	503	1005	1000 SAR		25-MAY-24	Debit Card	Yousef - Mada **** 8934
4	504	1006	1200 SAR		06-JUN-24	Credit Card	Ahmad - Mastercard **** 7741

```
SELECT Nationality, COUNT(*) AS Nationality_count FROM Guest GROUP BY Nationality HAVING COUNT(*) > 1;
```

Query Result x

SQL | All Rows Fetched: 1 in 0.002 seconds

NATIONALITY	NATIONALITY_COUNT
1 Saudi	3

5. Conclusion

In conclusion, The Hotel Booking and Management System successfully transformed disorganized and error-prone data handling into a structured and centralized database solution. The key outcome of our work is a measurable improvement in efficiency, data accuracy, and decision-making for hotel owners. By structuring reservations, room availability, payments, and services into clear data models; the system enables insightful analysis, such as tracking annual bookings to evaluate business growth and identifying attributes that impact success.

Throughout this project, our team gained strong database-related learning points, including: designing relational schemas, defining primary and foreign keys, enforcing entity relationships, writing optimized SQL queries, and understanding the importance of normalization to eliminate redundancy. We also learned that a database is not only about storing data, but about maintaining meaningful, real-world connections between entities to support business logic.

Our solution meets essential real-world database requirements by ensuring consistency, avoiding duplicate records, preserving referential integrity, and offering fast, query-based access to critical information. The system models realistic entities all which align directly with industry database standards, where structured data and clear relationships are vital for operational success.

Ultimately, our team solution fulfills core DBMS expectations by automating data storage and retrieval, reducing human error, and providing a scalable relational database architecture using SQL. This makes the system reliable, efficient, and fully aligned with the real-world needs of hotel booking and management operations.

Meeting Minutes

1st Meeting: 4 November

Through WhatsApp we agreed on a project domain to work on.

“A Hotel Booking & Management System”

2nd Meeting: 8 November

We had a read through the project together as a group finalizing our ideas to fully understand key concepts.

Required tasks for next meeting:

- Norah: find attributes for the entity Payment
- Lama: find the attributes for the Guest and staff entities
- Zainab: find attributes for the entity Rooms
- Shahad: find the attributes for the reservation and services entities

3rd Meeting: 10 November

We discussed what was required of us from the previous meeting. We then specified relationships, cardinalities, generalization/specialization relationships, and constraints.

Required tasks for next meeting:

- Shahad will hand draw the initial EER Diagram, we'll then review it as a team and add adjustments if needed
- Lama will begin sketching the schema

4th Meeting: 13 November

Finalized the EER diagram & Schema diagram drawings by reviewing them and fixing minor issues.

Required tasks for next meeting:

- Norah: Begin writing the introduction for our project + write SQL code for the entities Service and Guest.
- Zainab: write SQL code for the entity Staff.
- Lama: write SQL code for the entities Room and Payment.
- Shahad: write SQL code for the entity Reservation.

5th Meeting: 17 November

Through WhatsApp, each team member sent their created tables.

Required tasks for next meeting:

- Norah: write the SQL queries -> "Aggregation and grouping (GROUP BY, HAVING)"
- Zainab: write the SQL queries -> "Joins between multiple tables (INNER JOIN, LEFT JOIN)"
- Lama: write the SQL queries -> "Basic retrievals using SELECT-FROM-WHERE"
- Shahad: write the SQL queries -> "Sorting results in ascending/descending order"

6th Meeting: 30 November

We wrote the conclusion together and finalized the end report by reviewing it as a whole and fixing what needed to be fixed.